UHFusion: Prompt-Guided Multi-Task Learning Network for Hyperspectral Image Fusion via Multi-Type Mixture-of-Experts

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***Abstract*—Hyperspectral image (HSI) fusion aims to generate high-resolution HSI by integrating low-resolution hyperspectral data with auxiliary high-resolution sources (e.g., panchromatic (PAN), RGB, or MSI). While recent deep learning-based HSI fusion approaches have achieved promising results, they are typically designed for specific modality pairs and struggle to generalize across diverse fusion settings, leading to redundant architectures and limited adaptability. To overcome these limitations, we propose UHFusion, a prompt-guided multi-task learning (MTL) network for HSI fusion, which enables flexible multi-task adaptation within a single framework. Particularly, UHFusion introduces two key components: 1) a multi-type mixture-of-experts (MTMoEs) that decomposes HSI fusion into complementary spatial, spectral, and spatial–spectral reasoning processes, and dynamically composes expert responses conditioned on the auxiliary modality, allowing the model to selectively emphasize task-relevant spatial structures or spectral characteristics without architectural modification; and 2) a prompt-guided multi-task adaptation module (PMAM), which encodes modality-specific priors into learnable prompt embeddings and leverages a task relation graph to perform task-conditioned feature modulation, thereby improving the spatial–spectral fusion quality across diverse fusion tasks. Extensive experiments demonstrate that UHFusion achieves superior performance and generalization across multiple HSI fusion tasks and modalities, providing an effective all-in-one solution for HSI reconstruction. The code is available at: https://github.com/Jiahuiqu/UHFusion**

***Index Terms*—Hyperspectral classification, hyperspectral pansharpening, multi-task learning (MTL), mutual guidance.**

I. Introduction

HYPERSPECTRAL image (HSI) captures the radiation reflected from objects across contiguous spectral bands spanning the visible to near-infrared wavelengths [1], [2], [3], providing rich spectral information for remote sensing application, such as anomaly detection [4], [5], change detection [6], [7], and land cover classification [8], [9]. However, the intrinsic tradeoff between spectral and spatial resolution in optical imaging limits the spatial detail of HSIs, making raw HSIs insufficient for tasks requiring fine-grained spatial information. Therefore, the fusion of low-resolution HSIs (LR-HSIs) with auxiliary high-resolution panchromatic (PAN), multispectral (MSI), or RGB images has emerged as an effective strategy to generate high-resolution HSIs (HR-HSIs) with both fine spatial and rich spectral details.

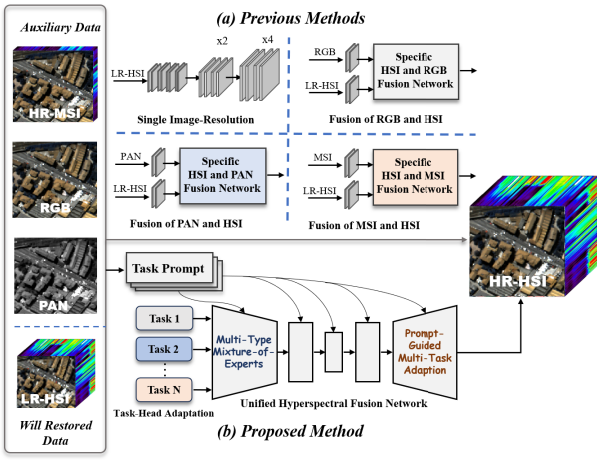


Fig. 1. Comparison between previous methods and the proposed multi-task HSI fusion framework. (a) Previous methods adopt separate task-specific architectures for different fusion settings, resulting in redundant designs and limited cross-task adaptability. (b) Proposed method employs a unified fusion network with MTMoE adapters and prompt-guided task modulation, enabling flexible adaptation to diverse fusion tasks.

With the rapid advancement of deep learning techniques [10], [11], [12], [13], [14], HSI fusion has achieved significant progress. As shown in Fig. 1, the existing HSI fusion methods are typically designed for a single task (e.g., HSI-PAN fusion [15], [16], HSI-MSI fusion [17], [18], or HSI-RGB fusion [19], [20]) and often rely on task-specific architectures or handcrafted priors. While these methods achieve desirable task-specific performance, the inherent task-specific design of these methods poses significant challenges for generalization and scalability to diverse modalities and real-world conditions. In practice, remote sensing systems frequently acquire auxiliary data from diverse sensors under complex, variable conditions, necessitating a unified and flexible fusion framework that can seamlessly adapt to multiple

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fusion settings without retraining modality-specific models. Some studies have explored unified frameworks for HSI fusion [21], [22]. However, most of these approaches simply integrate multiple task-specific fusion subnetworks in a modular or parallel manner to handle different inputs, without explicitly exploiting the shared characteristics across tasks. It often leads to redundant network structures and limited efficiency. Recently, the multi-task learning (MTL) paradigm has shown promise in low-level vision tasks [23], [24], demonstrating that prompt learning or conditional guidance can effectively enable a single model to handle multiple related tasks within a unified framework. Nevertheless, directly applying existing prompt-based strategies to HSI fusion remains suboptimal. Most prompt designs employ static or weakly conditioned task embeddings, which are insufficient to capture the large variations in resolution, spectral distribution, and modality-dependent degradation patterns across different fusion settings. As a result, these methods struggle to robustly adapt feature representations to diverse fusion objectives, particularly when the auxiliary modality changes significantly.

Based on mentioned earlier, we explore the MTL network for HSI fusion, aiming to develop a single model capable of dynamically adapting to various auxiliary inputs (e.g., PAN, RGB, MSI, or missing modalities) for high-quality HSI reconstruction. To this end, we propose UHFusion, which achieves multi-task adaptation within a unified framework by combining multi-type mixture-of-expert (MTMoE) and task-specific prompts, providing a flexible and generalizable solution for HSI reconstruction under diverse fusion conditions. Specifically, to fully exploit the complementary characteristics between HSI and arbitrary HR auxiliary sources in terms of spatial structures, spectral distributions, and spatial–spectral correlations, we propose the MTMoEs comprising dual-domain adaptive spatial MoE (DASMoE), multiscale frequency-aware spectral MoE (MFSMoE), and dynamic collaborative spatial–spectral MoE (DCS^{2}MoE). Equipped with a global-statistics-based gating mechanism, MTMoEs dynamically weight different experts according to the input characteristics, achieving flexible adaptation and fine-grained feature fusion across diverse modalities. During reconstruction, we propose the prompt-guided multi-task adaption module (PMAM), which encodes modality-specific priors into learnable prompt embeddings and integrates them via a scale-shift feature modulation mechanism, enabling the fusion network to adaptively guide multimodal features and enhance both its adaptability to diverse modalities and controllability across tasks. Overall, UHFusion enables flexible integration of multiple fusion tasks within a single unified architecture, offering a generalizable solution for HSI fusion under heterogeneous input conditions, and facilitating practical deployment in real-world remote sensing systems. Our contributions are as follows.

1) We propose a unified HSI fusion framework, termed as UHFusion, which flexibly accommodates diverse input conditions in an "all-in-one" manner, providing a generalizable framework for adaptive HSI fusion under diverse auxiliary modalities (e.g., PAN, RGB, MSI, and Missing situation).
2) An MTMoE module integrating spatial, spectral, and spatial–spectral MoEs is developed, which adaptively fuses multimodal features through frequency-domain compensation while balancing spatial detail and spectral fidelity.
3) We propose a PMAM that encodes modality-specific priors into learnable prompt embeddings and leverages a task relation graph to guide task-adaptive feature modulation, improving the spatial–spectral fusion quality across diverse fusion tasks.

The rest of this article is organized as follows. Section II presents the related work. Section III introduces the proposed UHFusion method. In Section IV, we conduct a series of experiments to evaluate the effectiveness of the proposed UHFusion. The conclusion is reported in Section V.

II. Related Work

A. HSI Fusion

Recently, numerous HSI fusion methods have been proposed, and the literature can be organized into four main families: component substitution (CS) methods [25], multiresolution analysis (MRA) methods [26], model-based methods [27], and deep learning-based methods [28], [29]. Generally, CS methods enhance spatial details effectively but often cause significant spectral distortion. MRA-based methods outperform CS-based methods in preserving spectral fidelity, but they are prone to artifacts and spatial distortion when the decomposition is improperly designed. Bayesian and matrix factorization approaches fall into the category of model-based methods. While they achieve satisfactory spatial–spectral preservation, their reliance on complex iterative optimization makes them unsuitable for large-scale or real-time applications. Deep learning-based methods can be further categorized into supervised and unsupervised approaches. Supervised methods [30], [31] generally achieve higher reconstruction quality by leveraging reference images as strong constraints, whereas unsupervised methods [32], [33] offer greater flexibility in scenarios without paired data. However, both paradigms typically adopt task-specific fusion strategies and have yet to realize a truly unified HSI fusion framework capable of handling arbitrary auxiliary data sources. In this work, we explore a novel approach to HSI fusion that leverages MTMoE along with lightweight, modality-specific prior prompt embeddings, enabling a unified model that can seamlessly integrate arbitrary auxiliary data sources, including scenarios with missing modalities.

B. MTL for HSI Fusion

MTL has emerged as an effective strategy for HSI processing, enabling shared representation learning across related tasks [34], [35]. In HSI fusion, MTL refers to jointly integrating HSIs with various high-resolution auxiliary modalities (e.g., PAN, RGB, and MSI) within a unified framework to support multiple fusion objectives. However, existing studies mainly focus on coupling upstream enhancement tasks with

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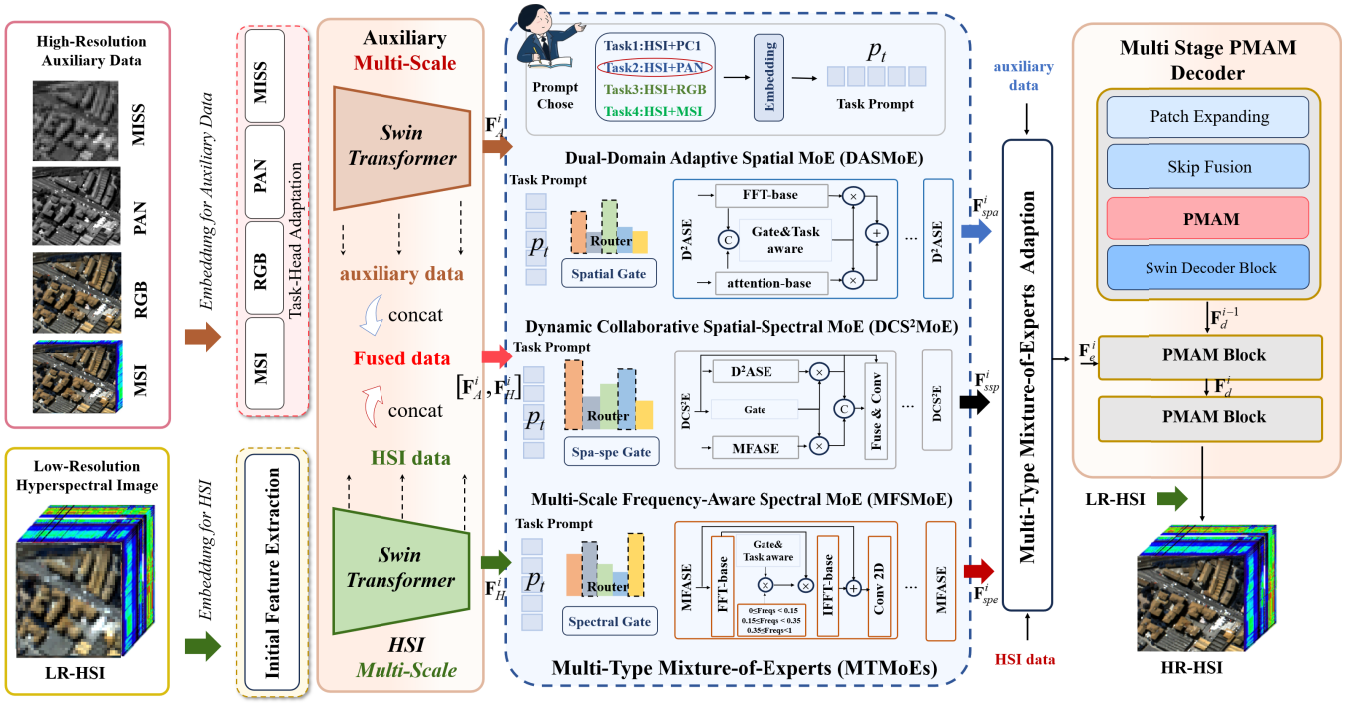


Fig. 2. Overall framework of the proposed UHFusion. The LR-HSI and high-resolution auxiliary modality (e.g., PAN, RGB, MSI, or PCA-derived guidance) are encoded by a dual-branch Swin-Transformer encoder to extract hierarchical HSI feature $\mathbf{F}_h$ and auxiliary-modality feature $\mathbf{F}_a$. At each scale, the proposed MTMoE adaptively integrates spatial, spectral, and spatial–spectral information via task prompt-based expert routing. The fused multiscale features are then progressively reconstructed by the shared decoder equipped with the prompt-guided PMAM, enabling task-aware feature modulation and HSI reconstruction across different fusion settings.

downstream vision tasks [36], [37], while unified multifusion-task learning remains largely unexplored. The limited efforts often rely on simplistic architectural compositions, leading to redundant parameters and weak task interaction. To address these issues, this work explores a unified MTL framework for HSI fusion through parameter sharing and task-aware prompting, facilitating more effective knowledge transfer and robust generalization across diverse fusion scenarios. Notably, the framework is capable of handling missing auxiliary modalities, ensuring broader applicability.

III. Proposed Method

A. Problem Formulation

In existing HSI fusion methods, different fusion tasks are usually modeled using task-specific networks. Given an LR-HSI $\mathbf{X}$ and a fixed high-resolution auxiliary modality $\mathbf{X}_{aux}^{(t)} \in \mathbb{R}^{c\times H\times W}$ corresponding to task t (e.g., PAN, RGB, or MSI), the fusion process is commonly formulated as

$$\widetilde{\mathbf{H}}^{(t)} = \Psi^{(t)}\left(\mathbf{X}, \mathbf{X}_{aux}^{(t)}\right) \tag{1}$$

where $\Psi^{(t)}(\cdot)$ denotes a task-specific fusion model trained independently for a particular auxiliary modality and fusion objective and $\widetilde{\mathbf{H}}^{(t)}$ denotes the reconstructed HR-HSI. However, such task-specific modeling strategies inherently limit flexibility and scalability, making it necessary to redesign network architectures or retrain separate models when auxiliary modalities or fusion configurations change.

To address these limitations, we reformulate HSI fusion as a unified MTL problem. Specifically, a single fusion network is learned to accommodate arbitrary auxiliary modalities by conditioning the fusion process on a task-specific prompt $\mathbf{p}_t$. The unified fusion formulation can be expressed as

$$\widetilde{\mathbf{H}} = \Psi\left(\mathbf{X}, \mathbf{X}_{aux}, \mathbf{p}_t; \theta\right) \tag{2}$$

where $\Psi(\cdot)$ represents a unified fusion model shared across different tasks and θ denotes the corresponding learnable parameters.

Within a unified HSI fusion framework, the key challenge lies in two aspects: 1) how to effectively model and integrate complementary spatial, spectral, and spatial–spectral information from heterogeneous modalities; and 2) how to flexibly adapt the reconstruction process to diverse fusion objectives with minimal additional complexity. As illustrated in Fig. 2, to address the challenges, we propose a unified HSI fusion framework, termed UHFusion, which integrates heterogeneous spatial–spectral information at each encoder scale through MTMoEs, while enabling flexible adaptation to diverse fusion objectives via a prompt-guided multi-task adaptation mechanism embedded in the shared decoder.

B. Multi-Task Integrated Swin-Unet for HSI Fusion

1) Dual-Branch Multiscale Features Encoding: The LR-HSI is first upsampled to match the spatial resolution of $\mathbf{X}_{aux}$ using bicubic interpolation and is denoted as $\mathbf{X}_u$. Subsequently, the proposed UHFusion projects these heterogeneous inputs into a unified latent subspace through two initial feature mapping layers, yielding the latent representations $\mathbf{F}_H^0$ and $\mathbf{F}_A^0$. These representations are then fed into a dual-branch multiscale encoder based on Swin-Transformer, where both the hyperspectral and auxiliary

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branches progressively extract hierarchical features across N scales (with N set to 3). The encoded features at the ith scale are denoted as $\mathbf{F}_H^i$ and $\mathbf{F}_A^i$, respectively. At each scale, the extracted spatial–spectral features are adaptively integrated through the MTMoEs mechanism, which jointly leverages spatial, spectral, and spatial–spectral MoEs to capture modality-specific cues and enhance cross-modal feature complementarity. The task prompt p_t is implemented as a discrete task identifier. In our experiments, $t = 0, 1, 2, 3$ corresponds to HSI and PAN fusion, HSI and RGB fusion, HSI and MSI fusion, and single HSI super-resolution (PCA-based), respectively. In MTMoEs, p_t is first mapped into a learnable prompt embedding, which is concatenated with the global pooled feature statistics to generate prompt-aware expert activation weights. Accordingly, $\mathbf{F}_A^i$ and p_t are fed into the DASMoE, $\mathbf{F}_H^i$ and p_t are fed into the MFSMoE, and their concatenated feature $[\mathbf{F}_A^i, \mathbf{F}_H^i]$ and p_t are fed into the DCS2MoE. The outputs of these three expert groups are then weighted and fused to obtain the reconstructed feature. The resulting fused representations at each scale can be formulated as

$$\mathbf{F}_e^i = \text{MTMoEs}\left(\mathbf{F}_H^i, \mathbf{F}_A^i; \mathbf{p}_t\right) = \mathbf{F}_{spa}^i + \mathbf{F}_{spe}^i + \mathbf{F}_{ssp}^i \tag{3}$$

where $\mathbf{F}_e^i$ denotes the fused feature at scale i; MTMoEs$(\cdot)$ denotes the MTMoEs, $\mathbf{p}_t$ denotes the task-specific prompt; and $\mathbf{F}_{spa}^i$, $\mathbf{F}_{spe}^i$, and $\mathbf{F}_{ssp}^i$ denote the outputs of the DASMoE, MFSMoE, and DCS^{2}MoE, respectively.

2) Task Prompt-Guided Shared Decoding: The shared decoder reconstructs the HR-HSI by progressively integrating multiscale fused features $\{\mathbf{F}_e^i\}_{i=1}^N$ extracted from the dual-branch encoder. To enable task-adaptive reconstruction across arbitrary auxiliary modalities, we proposed the PMAM, which dynamically modulates decoder features at each upsampling stage according to the fusion objective by the task-specific prompt $\mathbf{p}_t$. PMAM guides the decoder to selectively emphasize spatial or spectral cues relevant to the current task, thereby enhancing reconstruction while maintaining cross-modal complementarity. In PMAM, the input consists of the feature from MTMoEs and the same task prompt p_t. The feature is first globally averaged and projected into the prompt space to form a modality-aware token. Meanwhile, p_t is used to select the corresponding row of the learnable task-relation matrix, which aggregates task-specific prompt information from the task bank. The modality-aware token further queries the task bank to obtain a task-related prompt representation. These two prompt representations are combined and converted into channel-wise scale and shift parameters, which are then used to modulate the reconstructed feature in a FiLM-like manner. In this way, p_t explicitly guides both expert routing in MTMoEs and task-conditioned feature modulation in PMAM. Formally, let $\mathbf{F}_d^i$ denote the decoder feature at scale i, and $\mathbf{F}_e^i$ be the corresponding skip connection from the encoder. The decoding process can be expressed as

$$\mathbf{F}_d^i = \mathcal{U}^i\left(\mathcal{M}^i\left(\left[\mathbf{F}_d^{i-1}, \mathbf{F}_e^i\right], \mathbf{p}_t\right)\right) \tag{4}$$

where $[\cdot, \cdot]$ denotes channel-wise concatenation operation, $\mathcal{M}^i(\cdot, \mathbf{p}_t)$ represents the task-guided modulation at scale i, and $\mathcal{U}^i(\cdot)$ is the upsampling operator.

After the final decoding stage, a residual connection adds the upsampled LR-HSI $\mathbf{X}_u$ to the reconstructed features, which can be described by

$$\tilde{\mathbf{H}} = \mathbf{F}_d^N + \mathbf{X}_u. \tag{5}$$

By jointly leveraging the MTMoEs in the dual-branch encoder and PMAM in the shared decoder, UHFusion provides a unified framework for HSI fusion that can flexibly accommodate arbitrary auxiliary modalities. Moreover, when auxiliary data are unavailable, principal component analysis (PCA) is applied to the LR-HSI to extract salient spatial textures, effectively compensating for missing modalities and preserving consistent fusion performance.

C. Multi-Type Mixture of Experts for Spatial–Spectral Feature Fusion

HSI fusion within a unified multi-task framework faces the dual challenge of heterogeneous auxiliary modalities (e.g., PAN, RGB, and MSI) and diverse reconstruction objectives, including spatial enhancement and spectral fidelity. A simple and single-type feature extraction module is insufficient to simultaneously capture complementary cues across all modalities and tasks. Therefore, we propose the MTMoEs, which integrate frequency-domain transformations via fast Fourier transform (FFT) and comprise three synergistic MoEs: DASMoE, MFSMoE, and DCS^{2}MoE. Specifically, DASMoE focuses on spatial reasoning by exploiting high-resolution structural cues from the auxiliary modality, which mainly benefits texture and edge reconstruction. MFSMoE focuses on spectral reasoning by modeling band-wise dependencies and frequency-domain spectral responses from the HSI branch, which helps preserve spectral fidelity. In contrast, DCS^{2}MoE is designed to model coupled spatial–spectral interactions from the concatenated spatial–spectral representation. Therefore, DCS^{2}MoE is not a redundant branch overlapping with DASMoE or MFSMoE, but a complementary component that captures joint dependencies between spatial structures and spectral responses.

1) Dual-Domain Adaptive Spatial MoE: In HSI fusion, HR auxiliary modalities primarily contribute rich spatial structural and contextual information, which is essential for reconstructing fine spatial details and ensuring accurate spatial–spectral alignment. However, purely spatial-domain modeling often struggles to maintain global contextual consistency while preserving fine-grained textures. To this end, we design the DASMoE comprising multiple dual-domain adaptive spatial expert (D^{2}ASE), which enhances spatial contextual modeling through frequency-domain compensation. DASMoE enables adaptive reconstruction of spatial details while maintaining structural consistency across diverse auxiliary modalities and fusion tasks via dynamic expert routing. It can be written as

$$\mathbf{F}_{spa}^i = \sum_{j=1}^{N_{spa}} \alpha_{spa,j}^i \cdot \mathcal{E}_{spa,j}^i\left(\mathbf{F}_A^i, \mathbf{p}_t\right) \tag{6}$$

where $\mathcal{E}_{spa,j}^i(\cdot)$ denotes the jth D^{2}ASE and $\alpha_{spa,j}^i$ represents its corresponding attention weight. The weights are dynamically

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generated according to the global spatial statistics of the input features via a lightweight gating function

$$\boldsymbol{\alpha}_{spa}^{i} = \frac{\text{Softmax}\left(\mathbf{z}_{spa}^{i}\right) \odot \mathcal{M}_{\text{Top}-K}\left(\mathbf{z}_{spa}^{i}\right)}{\left\|\text{Softmax}\left(\mathbf{z}_{spa}^{i}\right) \odot \mathcal{M}_{\text{Top}-K}\left(\mathbf{z}_{spa}^{i}\right)\right\|_{1}} \tag{7}$$

in which

$$\mathbf{z}_{spa}^{i} = \mathcal{G}_{spa}^{i}\left(\left[\text{GAP}\left(\mathbf{F}_{A}^{i}\right), \mathbf{p}_{t}\right]\right) \tag{8}$$

and

$$\mathcal{M}_{\text{Top}-K}\left(\mathbf{z}_{j}\right) = \begin{cases} 1, & \mathrm{z}_{j} \in \text{Top} - \text{K}\,(z) \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where GAP$(\cdot)$ denotes global average pooling, $\mathcal{G}_{spa}^{i}(\cdot)$ is a learnable mapping function that projects the pooled feature to the expert weighting space, and $\mathcal{M}_{\text{Top}-K}(\cdot)$ represents the index of the $Top-K$ values to 1, and the rest to 0.

Each expert $\mathcal{E}_{spa,j}^{i}$ jointly models spatial dependencies and frequency-enhanced information through a dual-branch fusion process. In addition, DASMoE integrates a task-adaptive prompt $\mathbf{p}_t$ to guide each expert, enabling adaptive modulation of spatial and frequency features according to the fusion task, which enhances multi-task generalization and reconstruction fidelity. It can be expressed as

$$\mathcal{E}_{spa,j}^{i}\left(\mathbf{F}_{A}^{i}, \mathbf{p}_{t}\right) = f_{c}\left(\mathcal{G}_{\text{t}}\left(\mathcal{G}_{\text{g}}\left(\left[\mathcal{S}\left(\mathbf{F}_{A}^{i}\right),\ \mathcal{R}\left(\left|\mathcal{F}\left(\mathbf{F}_{A}^{i}\right)\right|\right)\right]\right), \mathbf{p}_{t}\right)\right) \tag{10}$$

in which

$$\mathcal{S}\left(\mathbf{F}_{A}^{i}\right) = f_{c}\left(\text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d}}\right)V\right) \tag{11}$$

$$\mathcal{R}\left(\left|\mathcal{F}\left(\mathbf{F}_{A}^{i}\right)\right|\right) = f_{c}\left(f_{dw}\left(\left|\mathcal{F}\left(\mathbf{F}_{A}^{i}\right)\right|\right)\right) \tag{12}$$

where $\mathcal{F}(\cdot)$ denotes the FFT operation; $\mathcal{G}_{\text{g}}(\cdot)$ and $\mathcal{G}_{\text{t}}(\cdot)$ denote the dual-domain gating and task-adaptive modulation, respectively; Q, K, and V are obtained via linear projections of $\mathbf{F}_{A}^{i}$; $f_{dw}(\cdot)$ is a depthwise convolution-based enhancement in the frequency domain; and $f_c(\cdot)$ is a convlolution layer with kernel size of 1×1.

2) Multiscale Frequency-Aware Spectral MoE: To ensure accurate spectral reconstruction in HSI fusion with arbitrary high-resolution auxiliary modalities, we design multiscale MFSMoE comprising multiple multiscale frequency-aware spectral experts (MFSEs), each specialized in capturing distinct pixel-level spectral patterns. Each expert splits the spectral channels into multiple scales, applies a 1-D FFT along the spectral dimension, learns adaptive frequency-domain masks via lightweight MLPs for each scale, and reconstructs enhanced spectral features via inverse FFT (IFFT). Task embeddings further modulate the spectral masks to adaptively emphasize the most relevant patterns for each fusion task. This design preserves spectral fidelity, suppresses noise, and produces robust spectral representations for cross-modal fusion. Formally, it can be written as follows:

$$\mathbf{F}_{spe}^{i} = \sum_{j=1}^{N_{spe}} \alpha_{spe,j}^{i} \cdot \mathcal{E}_{spe,j}^{i}\left(\mathbf{F}_{H}^{i}, \mathbf{p}_{t}\right) \tag{13}$$

where $\mathcal{E}_{ssp,j}^{i}(\cdot)$ denotes the jth multiscale frequency-aware spectral expert, $\alpha_{spe,j}^{i}$ is its adaptive weight computed via a lightweight gating mechanism, and p_i is the task prompt embedding.

Each expert operates via multiscale spectral modulation, with task-specific prompt embeddings providing adaptive guidance to modulate spectral responses according to the target fusion task. It can be written as

$$\mathcal{E}_{spe,j}^{i}\left(\mathbf{F}_{H}^{i}, \mathbf{p}_{t}\right) = f_{c}\Bigg(\mathbf{F}_{H}^{i} + \mathcal{F}^{-1}\Bigg[\mathcal{F}\left(\mathbf{F}_{H}^{i}\right) \odot\ \mathcal{G}_{t}\left(\mathbf{M}\left(\left|\mathcal{F}\left(\mathbf{F}_{H}^{i}\right)\right|\right), \mathbf{p}_{t}\right)\Bigg]\Bigg) \tag{14}$$

where $\mathbf{M}(|\mathcal{F}(\mathbf{F}_{H}^{i})|)$ denotes the multiscale spectral masks learned via lightweight MLPs on the FFT magnitude, $\mathcal{F}^{-1}[\cdot]$ denotes the IFFT, $\mathcal{G}_t(\cdot)$ represents the task-adaptive modulation applied to each scale, and $\odot$ denotes element-wise multiplication.

3) Dynamic Collaborative Spatial–Spectral MoE: Building upon the spatial and spectral experts, the DCS^{2}MoE comprising multiple dynamic collaborative spatial–spectral expert (DCS^{2}E) is designed to comprehensively model spatial–spectral dependencies within a unified representation space. Specifically, multiscale features $\mathbf{F}_{A}^{i}$ and $\mathbf{F}_{H}^{i}$ extracted from the dual-branch are hierarchically concatenated and fused to generate the composite representation $\mathbf{F}_{S}^{i}$, which serves as the input for joint feature enhancement. Each expert ensemble is responsible for capturing distinct spatial–spectral interaction patterns, while an adaptive gating mechanism dynamically modulates the contributions according to the intrinsic characteristics of the input data and the specific fusion objectives.

D. Prompt-Guided Multi-Task Adaption

To enable a single network to effectively handle diverse HSI fusion tasks, as shown in Fig. 3, we introduce the PMAM, which dynamically modulates decoder features using task-specific learnable prompts. By integrating task relation-guided feature sharing with lightweight channel-wise modulation, PMAM adaptively refines decoder representations across multiple tasks while preserving high spectral fidelity and spatial detail. It offers a simple yet powerful multi-task fusion solution, achieving task adaptability without incurring substantial computational overhead.

1) Relation-Guided Task Embedding With Attention Aggregation: PMAM leverages a learnable task bank relation graph to encode intertask dependencies and correlations. Through the relation matrix, embeddings of related tasks can share common representations, thereby enhancing feature generalization across heterogeneous fusion objectives. Let $\mathbf{T} \in \mathbb{R}^{N \times D}$ denote the learnable task bank, where N is the number of tasks and D is the embedding dimension. To capture both task-specific priors and intertask correlations, we construct a task relation-guided embedding $\mathbf{e}^{t}$ for input task $\mathbf{p}_t$ via a learnable relation graph $\mathbf{R} \in \mathbb{R}^{N \times N}$, followed by self-attention aggregation

$$\mathbf{e}^{t} = \sum_{k=1}^{N} \alpha_{k}^{t} \cdot \mathbf{T}^{k}, \quad \boldsymbol{\alpha}^{t} = \text{Softmax}\left(\mathbf{R}\left[\mathbf{p}_{t}, :\right]\right)$$

$$\mathbf{F}_{fused} = \sigma\left(\mathbf{W}_{c} \cdot \text{Softmax}\left(\frac{Q_{m}K_{t}^{\text{T}}}{\sqrt{d}}\right)V_{t}\right) + \lambda \cdot \mathbf{e}^{t} \tag{15}$$

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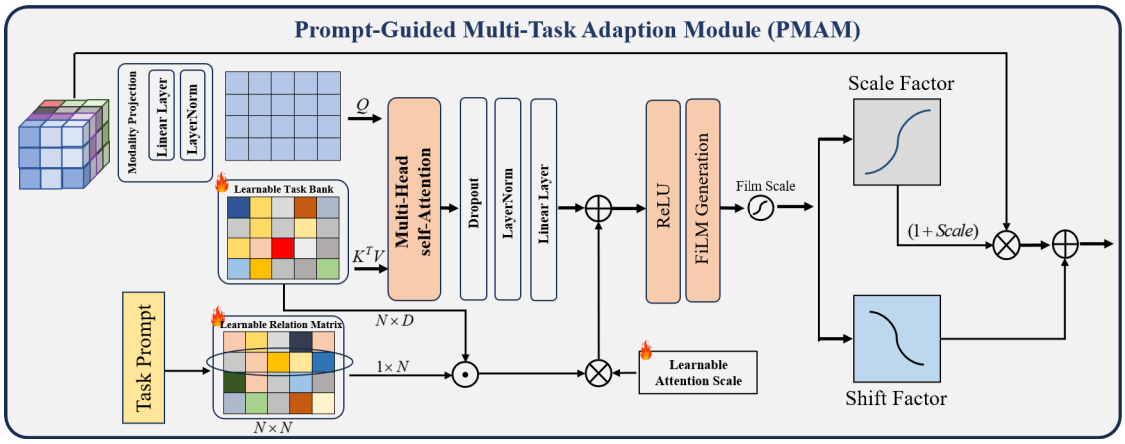


Fig. 3. Architecture of the proposed prompt-guided multi-task adaption module (PMAM). Given the decoder feature and task prompt p_t, PMAM first projects the feature into the prompt space as a modality-aware token. The token interacts with the learnable task bank through multihead attention, while the task relation matrix aggregates task-specific prompt information according to the current fusion task. The resulting prompt representation is further converted into channel-wise scale and shift parameters for FiLM-style feature modulation.

where α_k^t denotes the attention weight of task k on task t, Q_m is the query from the modality token, K_t and V_t are the key and value from the task bank $\mathbf{T}$, $\mathbf{W}_c$ is a learnable projection, and λ is the learnable scale modulation parameter.

2) Prompt-Guided Task-Adaptive Feature Modulation: In addition, PMAM employs a lightweight task-adaptive modulation mechanism to recalibrate decoder features. Specifically, the fused task embedding is mapped to a pair of scale and shift parameters $\boldsymbol{\gamma}^t$ and $\boldsymbol{\beta}^t$, which are applied to the decoder feature channels in a FiLM-style manner, transforming static decoder features into task-aware representations without adding significant complexity. It can be formulated as

$$\mathcal{M}(\mathbf{F}, \mathbf{p}_t) = \mathbf{F} \odot \left(1 + \boldsymbol{\gamma}^t\right) + \boldsymbol{\beta}^t$$
$$\left[\boldsymbol{\gamma}^t, \boldsymbol{\beta}^t\right] = \tanh\left(\mathbf{W} \cdot \mathbf{F}_{fused}\right) \cdot s \tag{16}$$

where $\mathbf{F}$ represents the decoding feature, $\mathbf{W}$ is a learnable projection, $\tanh(\cdot)$ is the hyperbolic tangent activation function, and s is a scale parameter, which is set to 0.1.

E. Loss Function

In this article, let $\{(\mathbf{X}_{hsi}^{(n)}, \mathbf{X}_{aux}^{(n)}, \mathbf{H}^{(n)})\}_{n=1}^{N}$ denote the training dataset, where N is the number of training samples. For the nth sample, $\mathbf{X}_{hsi}^{(n)}$ represents the input LR-HSI (after bicubic interpolation), $\mathbf{X}_{aux}^{(n)}$ denotes the corresponding high-resolution auxiliary modality (e.g., PAN, RGB, or MSI), and $\mathbf{H}^{(n)}$ is the ground-truth HR-HSI. The overall training objective is defined as

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \left\| \Psi\left(\mathbf{X}^{(n)}, \mathbf{X}_{aux}^{(n)}, \mathbf{p}_t; \theta\right) - \mathbf{H}^{(n)} \right\|_1. \tag{17}$$

IV. Experimental Results

A. Experiment Setting

To evaluate the performance of the proposed method, we conduct comprehensive experiments on two benchmark HSI datasets: Pavia Center [38] and Houston [39]. The Pavia Center dataset was acquired by the ROSIS sensor over Pavia, Italy, with a spatial resolution of 1.3 m. The original data contain 115 spectral bands covering the wavelength range from 0.4 to 0.9 μm, and 102 bands are typically retained for experimental analysis after the removal of 13 noisy bands affected by water vapor absorption. In our experiments, a representative region of 960 $\times$ 600 $\times$ 102 pixels is selected from the original image 1096 $\times$ 715. Then, we partition this region into 12 nonoverlapping subimages, which are then further cropped into 128 $\times$ 128 $\times$ 102 patches using the sliding window for generating the subsequent training and testing sample pairs.

Houston dataset was captured by the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding urban areas in Texas, USA. The dataset consists of 144 spectral bands covering the wavelength range from 0.38 to 1.05 μm, with a spatial resolution of 2.5 m per pixel. For our experiments, a subimage with the size of 320 $\times$ 1280 $\times$ 144 was cropped for evaluation of fusion performance. Similar to the Pavia Center dataset, it was also cropped to 128 $\times$ 128 $\times$ 144 using the sliding window for generating the subsequent training and testing sample pairs.

Following the commonly adopted Wald's protocol [40] and simulated spectral response function (SRF), we construct paired LR-HSIs and their corresponding high-resolution auxiliary data sources, including PAN, RGB, and MSI. The spatial resolution of the auxiliary modalities is set to 128 $\times$ 128, while that of the LR-HSI is 32 $\times$ 32. Each dataset is randomly divided into training and testing subsets with a 3:1 ratio.

The proposed UHFusion is implemented using the PyTorch framework and trained on two NVIDIA GeForce RTX 3090 GPUs. Specifically, the network is trained for 500 epochs using the Adam optimizer with an initial learning rate of $5e-4$ and a weight decay of $1e-6$. The batch size is set to 4, and the learning rate is reduced to one-tenth of its current value at epochs {100, 200, 250}. During training, the four fusion tasks, including HSI+PAN, HSI+RGB, HSI+MSI, and HSI+PCA, are sampled with equal probability. The same number of training samples is constructed for each task from the corresponding HR-HSI patches, thereby preventing a particular fusion setting from dominating the optimization. At each iteration, the selected samples are associated with their task identifiers and jointly optimized using the shared reconstruction loss.

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON TWO PUBLIC DATASETS. THE BEST AND SECOND-BEST PERFORMANCE ARE HIGHLIGHTED AND UNDERLINED

Dataset	Methods	HSI+PCA			HSI+PAN			HSI+RGB			HSI+MSI		
		PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓
Pavia Center	GSA	25.3610	6.8009	8.7168	29.8913	7.5829	4.9398	29.5347	6.4780	4.4389	33.5273	5.6095	3.1716
	CNMF	25.2528	6.8909	8.8129	27.4760	7.0037	6.5082	30.1199	5.6087	4.5776	34.2195	4.1599	3.4229
	PANC	27.5862	6.2532	6.9600	35.9326	3.6995	2.3648	37.9652	3.2385	1.9102	40.3121	3.9422	1.9574
	ESFS	28.1493	5.2585	6.5184	36.5863	4.0357	2.3026	37.6655	3.5712	2.0297	40.5228	3.4927	1.7865
	BPDUN	28.0231	6.5519	6.5660	36.7632	3.8634	2.2199	37.4276	3.3087	1.9485	40.1543	2.9970	1.5530
	MSAN	26.4163	9.3851	7.8054	35.9081	4.0329	2.4003	37.4703	3.0531	1.8884	42.5656	2.8916	1.5076
	S3FNet	28.0234	5.5844	6.5830	36.7809	3.6918	2.2303	37.9736	3.6661	1.9844	41.5307	3.7396	1.7269
	PANM	27.3940	5.6888	7.1118	37.2286	3.5104	2.1419	38.9212	3.1291	1.7824	42.4486	2.8689	1.5520
	SFIGNet	26.8880	6.1553	7.5893	34.1433	4.7066	2.8758	37.9436	2.9424	1.7749	42.9674	3.0089	1.5556
	AELF	27.9616	5.9515	6.6802	35.9591	4.1333	2.4114	38.8724	2.9015	1.6913	42.6373	2.7700	1.5668
	IFPMH	28.1389	5.4942	6.6861	36.7455	3.5633	2.3630	38.3139	2.8565	2.0136	40.4737	2.4394	1.7495
	UniNet	29.5520	4.8329	5.6122	36.9907	3.5179	2.2698	37.6813	3.1207	2.0554	40.8143	2.4927	1.6711
	Ours	**30.6279**	**4.6091**	**5.0117**	**38.2439**	**3.3201**	**1.9784**	**39.5450**	**2.7714**	**1.6855**	**43.2074**	**2.4684**	**1.4039**
Houston	GSA	29.4341	6.3049	7.2665	32.7198	9.5454	5.3919	35.1198	5.1679	3.8108	40.2124	3.4811	2.2444
	CNMF	29.3257	6.7385	7.3817	33.4132	5.7701	4.7003	36.4948	5.0940	3.3809	39.2085	3.8101	2.7144
	PANC	31.7308	5.6359	5.7102	39.1067	3.0506	2.4673	43.4884	1.8487	1.6048	48.4331	1.3676	1.0979
	ESFS	31.6826	4.9375	5.6457	40.2218	2.7184	2.2071	41.7917	2.0300	1.8310	48.6422	1.4560	1.2080
	BPDUN	31.9402	5.3569	5.7326	39.6161	3.0389	2.4479	43.1928	1.9827	1.8278	47.8306	1.2980	1.1809
	MSAN	30.3763	5.8879	6.5528	38.7605	3.1615	2.5612	40.1387	2.4087	2.1814	48.3298	1.2951	1.0610
	S3FNet	31.5984	5.1868	5.7059	39.7619	2.9197	2.2973	43.9518	1.8942	1.5892	48.2726	1.6134	1.1851
	PANM	30.8006	5.5746	6.2151	39.6928	2.9107	2.3292	43.6233	1.8795	1.5552	48.6610	1.2135	1.0225
	SFIGNet	30.3261	5.4971	6.6102	34.1433	4.7066	2.8758	37.9436	2.9424	1.7749	42.9674	3.0089	1.5556
	AELF	31.8471	5.3197	5.5796	38.4418	3.5848	2.7218	39.5683	2.8314	2.3750	49.1122	1.3759	1.0818
	IFPMH	29.7085	6.5672	7.1780	39.4883	2.6997	2.3491	40.0471	2.3606	2.2271	48.0274	1.2855	1.0702
	UniNet	29.1768	6.5954	7.5511	38.9122	2.8871	2.4977	41.8946	1.8486	1.8136	50.3359	1.1368	0.9802
	Ours	**33.2841**	**3.9596**	**4.7001**	**41.2254**	**2.4193**	**1.9645**	**44.8962**	**1.5834**	**1.3531**	**50.5844**	**1.0029**	**0.8695**

B. Comparison With State-of-the-Art Methods

In this article, we compare the performance of the proposed UHFusion with various both classical and the state-of-the-art HSI fusion methods, including GSA [25], CNMF [27], PANC [29], ESFS [30], BPDUN [41], MSAN [42], S3FNet [43], PANM [44], SFIGNet [45], AELF [17], IFPMH [22], and UniNet [21]. To evaluate fusion performance, we use both qualitative evaluation through visual inspection and three quantitatively evaluated metrics. Specifically, we visualize the reconstructed images and their corresponding spatial and spectral error maps. These error maps highlight the local reconstruction fidelity, where a lower intensity indicates a higher similarity to the reference image. In addition, three commonly used metrics, peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), and erreur relative globale adimensionnelle de synthése (ERGAS), are adopted for quantitative evaluation.

1) Quantitative Performance Analysis: Table I reports the comprehensive quantitative comparison of various fusion methods on the Pavia Center and Houston datasets under four distinct fusion settings (HSI+PCA, HSI+PAN, HSI+RGB, and HSI+MSI). It can be observed that the traditional methods, CNMF and GSA, suffer from severe spatial and spectral distortion compared with the deep learning-based approaches, as indicated by their higher SAM and ERGAS values. In contrast, deep learning-based approaches show a substantial leap in reconstruction quality. Although PANC, BPDUN, MSAN, SFIGNet, and AELF achieve relatively competitive results in certain metrics, their overall performance remains suboptimal due to limited exploitation of spatial–spectral complementary features. ESFS and PANM demonstrate relatively balanced performance, yet they still exhibit lower PSNR and higher SAM compared to the proposed UHFusion. The recently developed unified and integrated methods, UniNet and IFPMH, also achieve competitive results in several settings, such as UniNet on the HSI+MSI fusion task of the Houston dataset. However, their performance is less consistent across different auxiliary modalities. In contrast, UHFusion consistently delivers the best results across all modalities and evaluation metrics, significantly outperforming existing task-specific fusion methods. Notably, the substantial gains under the HSI+PCA and HSI+MSI settings demonstrate its strong robustness to missing or heterogeneous auxiliary inputs.

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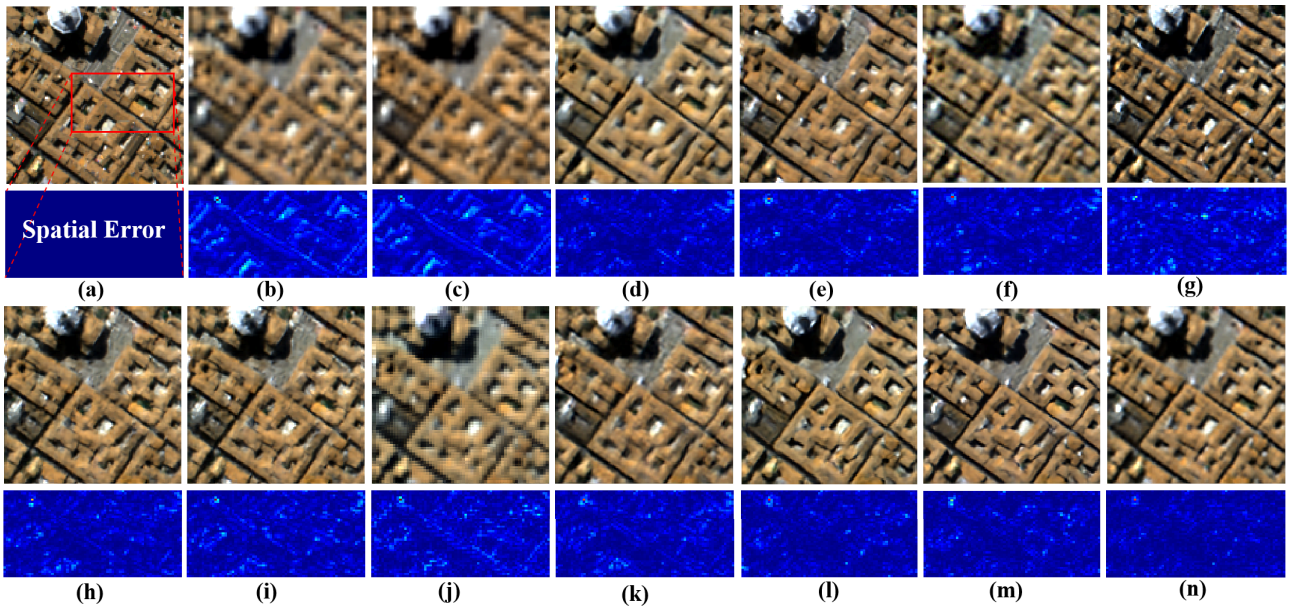


Fig. 4. First rows of each method: the results for the task of fusion using PC1 as auxiliary modality (pseudo-color band: 20–40–60) on the Pavia Center dataset. Second row of each method: corresponding spatial errors (blue means small differences and yellow means large differences). (a) Reference. (b) GSA. (c) CNMF. (d) PANC. (e) ESFS. (f) BPDUN. (g) MSAN. (h) S3FNet. (i) PANM. (j) SFIGNet. (k) AELF. (l) IFPMH. (m) UniNet. (n) Proposed.

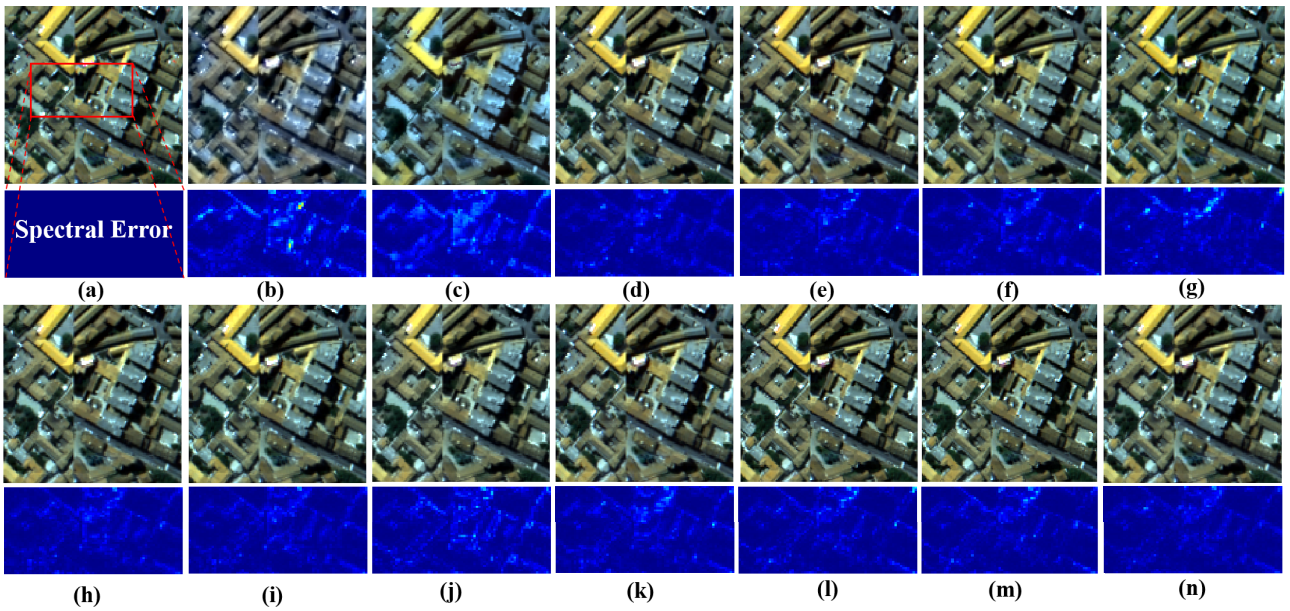


Fig. 5. First rows of each method: the results for the task of fusion using PAN as auxiliary modality (pseudo-color band: 20–40–60) on the Pavia Center dataset. Second row of each method: corresponding spectral errors (blue means small differences and yellow means large differences). (a) Reference. (b) GSA. (c) CNMF. (d) PANC. (e) ESFS. (f) BPDUN. (g) MSAN. (h) S3FNet. (i) PANM. (j) SFIGNet. (k) AELF. (l) IFPMH. (m) UniNet. (n) Proposed.

This superiority can be attributed to the proposed unified MTL paradigm, which enables shared representation learning across different fusion settings and facilitates the transfer of complementary spatial–spectral knowledge among four fusion tasks.

2) Visual Comparison of Reconstructed Results: Figs. 4–7 illustrate the visual comparison of different fusion methods under four distinct configurations: HSI+PCA, HSI+PAN, HSI+RGB, and HSI+MSI. In each figure, the first and third rows present the reconstructed HR-HSIs generated by all competing methods, while the second and fourth rows display the corresponding spatial and spectral error maps. It can be seen that the classical GSA and CNMF methods suffer from severe distortions in both reconstruction quality and spatial–spectral consistency, which aligns with the inferior quantitative scores. The deep learning-based methods, including PANC, ESFS, BPDUN, MSAN, PANM, and AELF, generate visually reconstructed results close to the reference image. However, their spatial and spectral error maps reveal noticeable artifacts in texture-rich regions, such as building edges, indicating limited enhancement of spatial details and insufficient preservation of spectral fidelity. The unified and integrated methods, UniNet and IFPMH, still show noticeable reconstruction errors around texture-rich boundaries and structural edges, with relatively larger responses in the corresponding spatial and spectral error maps, indicating that there remains room for

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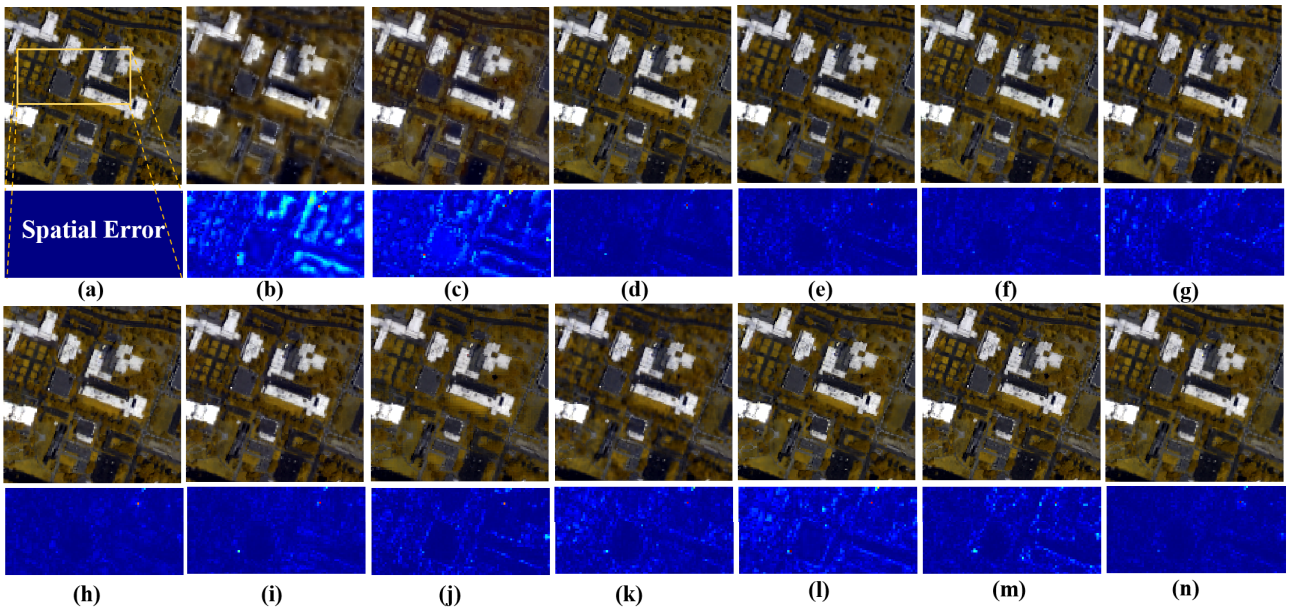


Fig. 6. First rows of each method: the results for the task of fusion using RGB as auxiliary modality (pseudo-color band: 50–80–90) on the Houston dataset. Second row of each method: corresponding spatial errors (blue means small differences and yellow means large differences). (a) Reference. (b) GSA. (c) CNMF. (d) PANC. (e) ESFS. (f) BPDUN. (g) MSAN. (h) S3FNet. (i) PANM. (j) SFIGNet. (k) AELF. (l) IFPMH. (m) UniNet. (n) Proposed.

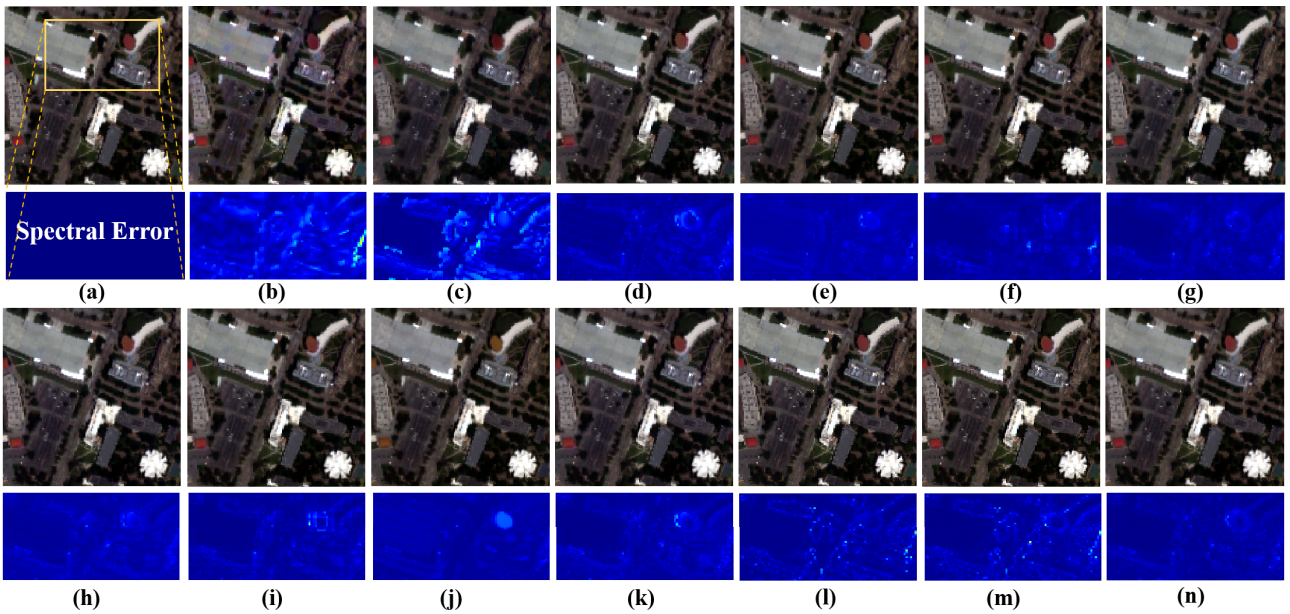


Fig. 7. First rows of each method: the results for the task of fusion using MSI as auxiliary modality (pseudo-color band: 20–40–60) on the Houston dataset. Second row of each method: corresponding spatial errors (blue means small differences and yellow means large differences). (a) Reference. (b) GSA. (c) CNMF. (d) PANC. (e) ESFS. (f) BPDUN. (g) MSAN. (h) S3FNet. (i) PANM. (j) SFIGNet. (k) AELF. (l) IFPMH. (m) UniNet. (n) Proposed.

improvement in preserving fine spatial details and spectral consistency across heterogeneous inputs. The proposed UHFusion exhibits error maps that are more consistent with those of the reference image, demonstrating its superior ability to balance spatial resolution and spectral fidelity across different fusion conditions. This advantage mainly benefits from the collaborative design of MTMoEs and PMAM. Specifically, MTMoEs decomposes the fusion process into spatial, spectral, and spatial–spectral expert reasoning, where DASMoE enhances high-resolution spatial structures from the auxiliary modality, MFSMoE preserves spectral dependencies from the HSI branch, and DCS2MoE further captures their coupled spatial–spectral interactions. Meanwhile, PMAM enables the reconstruction process to be adaptively adjusted for different fusion objectives.

3) Testing Efficiency: We evaluate the computational complexity of different methods in terms of parameter count, FLOPs, and inference time. The experiments are conducted on the HSI+PAN fusion task of the Pavia Center dataset. GSA, CNMF, and SFIGNet are executed on an Intel Core i5-10400F CPU @ 2.9 GHz, while the remaining deep learning-based methods are evaluated on an NVIDIA RTX 3090 GPU. As shown in Table II, it can be seen that although some competing methods have fewer parameters or shorter inference time,

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TABLE II
COMPUTATIONAL COST AND INFERENCE TIME OF PAN-HSI FUSION METHODS ON THE PAVIA CENTER DATASET

Method	GSA	CNMF	SFIGNet	ESFS	BPDUN	MSAN	S^{3}FNet	PANM	AELF	IFPMH	UniNet	Proposed
Params (M)	–	–	–	7.91	0.89	6.05	1.58	0.26	6.93	30.40	31.82	28.81
FLOPs (G)	–	–	–	215.01	13.21	64.27	9.66	4.29	33.22	291.32	518.03	7.42
Inference time (s)	0.160	0.390	0.057	0.080	0.035	0.008	0.026	0.023	0.025	0.061	0.077	0.069

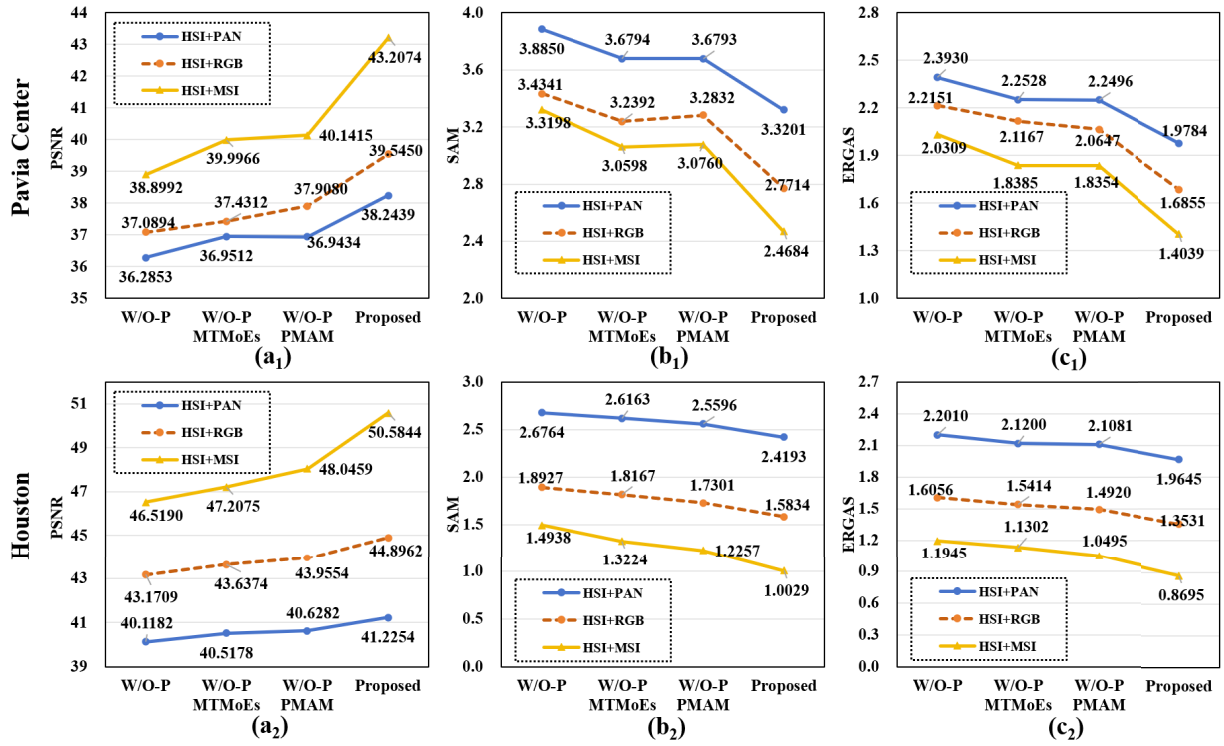


Fig. 8. Comparative analysis of fusion performance for task prompt in different modules. (a1)–(c1) and (a2)–(c2) Show the fusion performance in terms of PSNR, SAM, and ERGAS on the Pavia Center and Houston datasets, respectively.

they generally achieve lower reconstruction accuracy across different fusion settings. UHFusion has more parameters than several lightweight fusion models because it jointly handles multiple fusion tasks within a unified framework. A similar tradeoff is also observed in the unified and integrated methods. Nevertheless, UHFusion reduces model redundancy compared with these unified methods while maintaining adaptation to diverse fusion settings. In addition, the Top-K routing mechanism activates only a subset of experts for each input, thereby controlling the actual computational cost.

C. Ablation Study

1) Influence of Task Prompt in Different Modules: Task prompt serves as an essential component of the proposed framework by enabling explicit task-aware conditioning within a unified fusion architecture. To quantitatively evaluate the contribution of the prompt-guided design, we compare the proposed method with three degraded variants: 1) a version without prompt embeddings (denoted as W/O-P); 2) a variant where prompt embeddings are removed from the MTMoEs (W/O-P MTMoEs); and 3) a variant where prompt embeddings are excluded from the PMAM (W/O-P PMAM). As reported in Fig. 8, the removal of prompt embeddings consistently leads to performance degradation across all fusion tasks, with more evident drops observed in the fusion of HSI and MSI on PSNR. These results indicate that prompt-guided learning effectively facilitates task-aware feature adaptation.

2) Influence of Multi-Type Expert Variants: To evaluate the effectiveness of the MTMoEs, we compare the full MTMoEs with a baseline model without any MoE module, as well as four expert variants: SpaMoE with only spatial experts, SpeMoE with only spectral experts, SSpMoE with only spatial–spectral experts, and the combination of SpaMoE and SpeMoE. As shown in Table III, SpaMoE and SpeMoE achieve comparable performance, indicating that modeling spatial or spectral information alone is insufficient to fully capture complex cross-modal complementarities. The combination of SpaMoE and SpeMoE achieves the second-best performance by jointly modeling spatial and spectral features. In contrast, the proposed MTMoEs consistently deliver the best overall performance.

3) Influence of Expert Numbers in MTMoEs: To investigate the expert configuration of MTMoEs, we first evaluate the number of subexperts in each expert branch. MTMoEs consists of three branches, i.e., DASMoE, MFSMoE, and DCS2MoE, each containing multiple subexperts with identical

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TABLE III
QUANTITATIVE COMPARISON OF THE INFLUENCE OF MULTI-TYPE EXPERT VARIANTS ON TWO PUBLIC DATASETS

Dataset	Configuration			HSI+PAN			HSI+RGB			HSI+MSI		
	SpaMoE	SpeMoE	SSpMoE	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓
Pavia Center	×	×	×	35.1257	4.6559	2.8083	35.3650	4.4950	2.7743	36.9201	4.1004	2.5196
	✓	×	×	35.8191	3.9843	2.5173	36.6069	3.6115	2.3452	38.5458	3.4045	2.1230
	×	✓	×	35.7240	4.1924	2.5446	36.0782	4.1274	2.4581	38.4570	3.6176	2.1686
	×	×	✓	36.8005	3.7181	2.2672	37.8305	3.2635	2.0574	40.4784	3.0095	1.7817
	✓	✓	×	37.2354	3.5474	2.1863	37.7882	3.0954	2.0382	40.8094	2.8604	1.7194
	✓	✓	✓	**38.2439**	**3.3201**	**1.9784**	**39.5450**	**2.7714**	**1.6855**	**43.2074**	**2.4684**	**1.4039**
Houston	×	×	×	39.4732	2.9905	2.3813	40.6573	2.6044	2.1135	43.1595	2.1431	1.6552
	✓	×	×	39.8950	2.7820	2.2854	42.2297	2.1104	1.7882	45.0545	1.6310	1.3817
	×	✓	×	39.6873	2.8938	2.3603	42.3597	2.1476	1.8008	44.7794	1.8059	1.4508
	×	×	✓	39.9816	2.7620	2.2338	42.9027	1.9755	1.6486	46.1615	1.4317	1.2234
	✓	✓	×	40.5110	2.5678	2.1145	43.6568	1.7606	1.5304	46.9920	1.2922	1.1411
	✓	✓	✓	**41.2254**	**2.4193**	**1.9645**	**44.8962**	**1.5834**	**1.3531**	**50.5844**	**1.0029**	**0.8695**

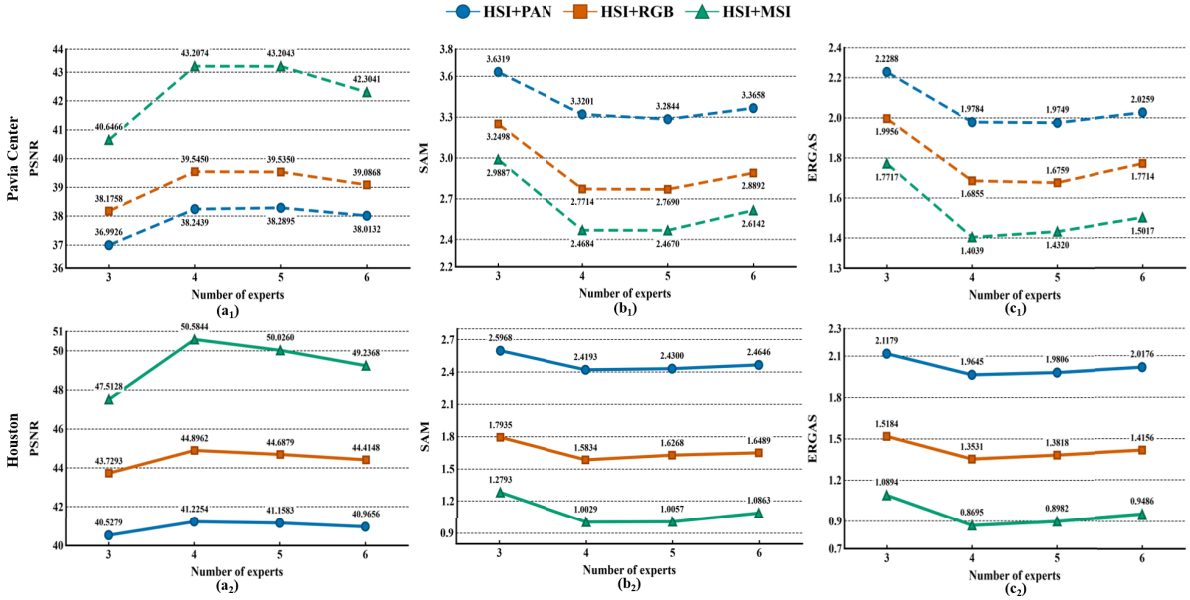


Fig. 9. Quantitative metrics comparison results for the influence of the number of experts in MTMoEs. (a1)–(c1) and (a2)–(c2) Show the fusion performance in terms of PSNR, SAM, and ERGAS on the Pavia Center and Houston datasets, respectively.

architectures but different parameters. As shown in Fig. 9, using four subexperts in each branch achieves the best overall performance. Therefore, the number of subexperts in DASMoE, MFSMoE, and DCS2MoE is fixed to 4.

4) Influence of Top-K Routing in MTMoEs: We further investigate the Top-K routing strategy by varying the number of activated subexperts within each branch, i.e., $K \in \{1, 2, 3, 4\}$. As shown in Fig. 10, increasing K from 1 to 2 clearly improves the reconstruction performance, whereas activating more than two experts brings only marginal gains with increased computational cost. Therefore, $K = 2$ is selected as the default setting. To maintain a unified routing configuration and avoid task-specific hyperparameter tuning, the same K is adopted for all fusion tasks. It should be noted that the same K only constrains the number of activated experts rather than their identities. Since the routing scores are jointly determined by the task prompt and the global statistics of the input features, different tasks and input samples can activate different subexperts with different aggregation weights. In this way, a common routing sparsity is maintained while task- and content-adaptive expert selection is preserved across different fusion settings.

5) Influence of Different Task Interference in Unified Network: To further examine task interference, we compared the jointly trained UHFusion with four task-specific variants trained independently using the same backbone and experimental settings. As shown in Table IV, joint training does not cause a consistent performance degradation on any particular modality and achieves comparable or improved reconstruction performance across the four fusion settings. This indicates that the shared representation learning among related HSI fusion tasks provides complementary spatial–spectral knowledge rather than producing severe neg-

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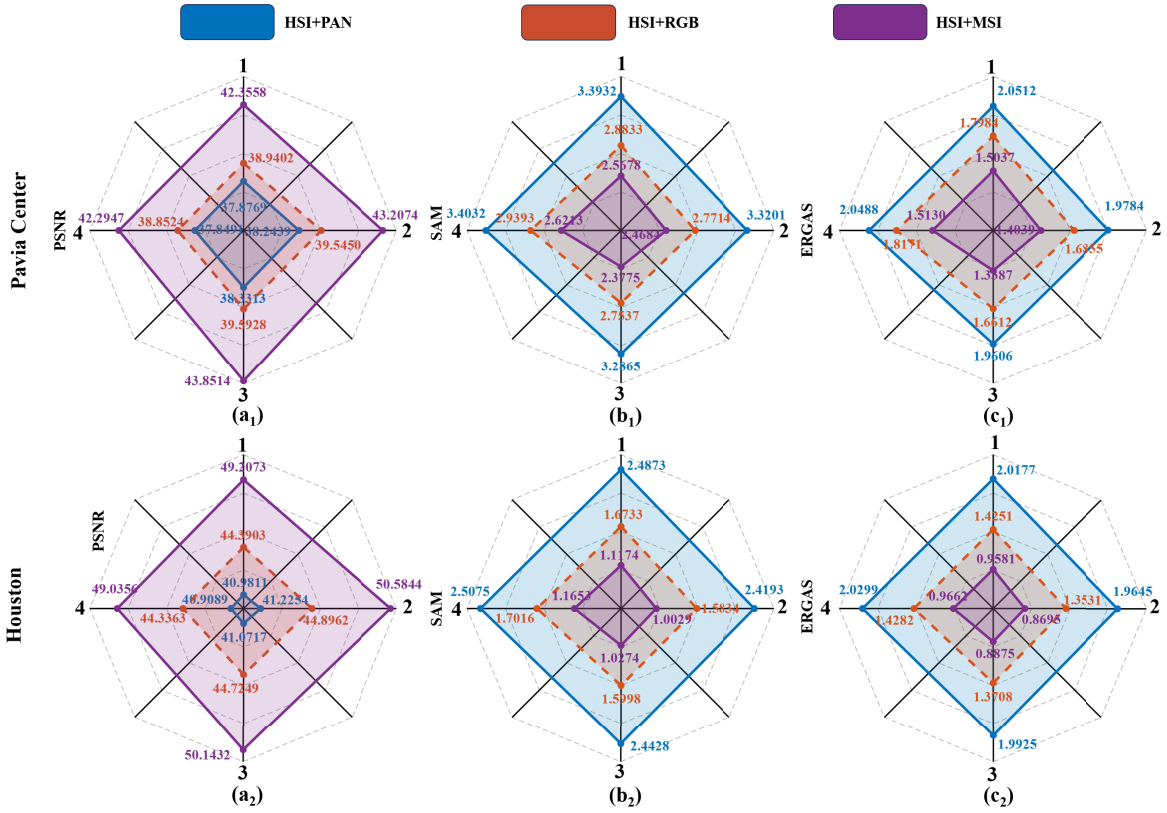


Fig. 10. Quantitative metrics comparison results for the influence of the number of activated experts in MTMoEs. (a1)–(c1) and (a2)–(c2) Show the fusion performance in terms of PSNR, SAM, and ERGAS on the Pavia Center and Houston datasets, respectively.

TABLE IV

QUANTITATIVE COMPARISON BETWEEN JOINT MULTI-TASK TRAINING AND TASK-SPECIFIC TRAINING ON TWO PUBLIC DATASETS

Dataset	Task	HSI+PCA			HSI+PAN			HSI+RGB			HSI+MSI		
		PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓
Pavia Center	Single	30.4418	4.8155	5.1312	38.0774	3.4109	2.0178	39.2467	2.8241	1.7431	42.9209	2.6313	1.5029
	Unified	**30.6279**	**4.6091**	**5.0117**	**38.2439**	**3.3201**	**1.9784**	**39.5450**	**2.7714**	**1.6855**	**43.2074**	**2.4684**	**1.4039**
Houston	Single	31.2575	5.2129	6.2632	40.8110	2.4906	2.0465	44.6959	1.5919	1.3913	50.2899	1.0385	0.8956
	Unified	**33.2841**	**3.9596**	**4.7001**	**41.2254**	**2.4193**	**1.9645**	**44.8962**	**1.5834**	**1.3531**	**50.5844**	**1.0029**	**0.8695**

ative transfer. In addition, the task-conditioned expert routing in MTMoEs and feature modulation in PMAM allow different fusion tasks to selectively activate and emphasize task-relevant representations, further reducing interference among heterogeneous modalities.

V. CONCLUSION

In this article, we present a prompt-guided MTL framework for HSI fusion, which enables a single network to flexibly adapt to diverse fusion tasks and heterogeneous high-resolution auxiliary modalities. Specifically, we introduce an MTMoE architecture consisting of spatial, spectral, and spatial–spectral experts, which collaboratively model complementary information across modalities and feature domains through dynamic expert activation and weighting. MTMoEs allows to selectively emphasize task-relevant spatial structures, spectral distributions, and their correlations under varying fusion conditions. In addition, the proposed PMAM injects task-specific priors into the reconstruction process via lightweight scale-shift feature modulation, enabling effective task-aware guidance without introducing significant computational overhead. As a result, the proposed framework achieves robust and consistent performance across multiple HSI fusion scenarios. In future work, we will further investigate how to extend this unified prompt-guided expert collaboration framework to a broader range of HSI quality enhancement tasks, such as HSI denoising, deblurring, destriping, and restoration under different degradation conditions. In addition, we will explore more compact expert-sharing and expert-pruning strategies to improve the computational efficiency and scalability of the multi-type MoE architecture.

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